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Smishing risk assessment using RAG and its potential for pedagogical recontextualization for cybersecurity education

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ABSTRACT

SMS phishing ("smishing") is an escalating threat that exploits mobile messaging to deliver malicious links. We present an explainable retrieval-augmented generation (RAG) prototype that flags smishing in Lithuanian SMS and explains its reasoning. The system extracts text from screenshots and evaluates each message against five criteria: search engine indexing, domain allow listing, use of a URL shortener, matches in a phishing database, and semantic similarity to labeled messages. An LLM dynamically weights these factors, generates a risk score ranging from 0 to 100, and provides a natural-language justification.

To develop cybersecurity literacy for recognizing threats in mobile messaging environments, it is essential first to understand the technical mechanisms that enable risk identification and to recontextualize those methods in an educational context. The prototype was tested on 20 SMS messages with three LLMs drawn from three distinct providers - OpenAI GPT o1, Anthropic Claude Sonnet 3.7, and DeepSeek V3 - evaluated under an identical pipeline. Claude Sonnet delivered the most stable scores (mean coefficient of variation ≈ 0.10). Semantic similarity had the highest cross-model average weight (23.5%), ranking first or second across the five criteria for every model tested, suggesting generalizability beyond a single model family.

Results indicate that the RAG approach combines reliable detection with transparent feedback, enabling users to see why a message is risky. By coupling technical accuracy with interpretable explanations, the tool has the potential to function not only as a smishing filter but also to be pedagogically recontextualized as a supportive resource for fostering cybersecurity literacy.

1. Introduction

Over the past decade, phishing has emerged as a significant and alarming issue, with numerous studies analyzing attack patterns and forecasting the use of various techniques (Abdillah et al., 2022; Ali & Mohd Zaharon, 2022; Varshney et al., 2016). SMS phishing (smishing) is a critical and evolving cybersecurity challenge that leverages the immediacy of mobile messaging to deceive users into disclosing sensitive data or clicking on malicious links. Smishing closely resembles email-based attacks, aiming to obtain the victim's credentials. Typically, attackers send text messages that include either a phone number intended for follow-up communication or a URL that directs the recipient to a fraudulent website (Mishra & Soni, 2020). Phishing attacks rose to 989,123 in Q4 2024, driven by a surge in SMS scams sent by Chinese actors using a new phishing kit and TOP domains (APWG, 2025).

Despite widespread awareness of online scams, studies show that users often fail to exercise caution when receiving SMS messages. For instance, Edwards et al. (2023) found that although many individuals recognized the theoretical risks of clicking on unknown links, a majority still engaged with them in practice, indicating a significant gap between awareness and behavior. Kumarasinghe et al. (2023) reported similar results. Although several anti-smishing tools have been developed to block or filter such SMS, research shows that they are not effective enough (Timko & Rahman, 2023). This underscores the need for tools that not only detect smishing but also educate users on how to identify and avoid it.

Traditional phishing detection systems are typically opaque, offering limited insight into the reasoning behind their decisions (Lim et al., 2025). Such "black box" models are ill-suited for educational contexts where explainability and user engagement are essential (Bhagat et al., 2023). In contrast, explainable artificial intelligence (XAI) frameworks offer transparent reasoning processes that support learning outcomes, foster user trust, and facilitate critical reflection on automated decisions (Melo et al., 2022). A structured literature search (Scopus; keywords

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"smishing OR phishing", "Retrieval-Augmented Generation") did not identify prior peer-reviewed studies addressing explainable RAG-based smishing detection in an explicitly educational context. This study therefore contributes a novel RAG-based, explainable smishing detection framework, designed as a technically validated foundation for future recontextualization and integration in an educational context to support the development of cybersecurity literacy.

While multimedia cybersecurity tools, such as educational games or simulations, have shown promise in increasing awareness, many lack robust evaluations of their effectiveness, especially in smishing-specific contexts (Zhang-Kennedy & Chiasson, 2021). Teachers also report a lack of accessible, interpretable tools for cybersecurity education, with existing materials often being fragmented or overly technical (McLeod et al., 2024). Therefore, it is important to identify options for pedagogical recontextualization of other materials that can be used in an educational context.

An additional gap concerns the linguistic and regional specificity of smishing detection tools. Most existing phishing and smishing detection frameworks have been developed and evaluated primarily on English-language datasets or large multilingual corpora dominated by high-resource languages (Dunđer et al., 2023; Salloum et al., 2023). However, phishing messages often exploit language-specific stylistic patterns, grammatical constructions, and culturally contextual cues that may not generalize across linguistic environments. In smaller language communities such as Lithuanian, the limited availability of labeled datasets and domain-specific corpora further constrains the effectiveness of traditional supervised approaches. As a result, detection systems optimized for English or other high-resource languages may underperform on Lithuanian SMS messages, particularly in identifying nuanced semantic manipulation tactics. Addressing this limitation requires approaches that leverage contextual similarity and dynamic knowledge retrieval rather than relying solely on static keyword patterns or large-scale labeled corpora. The combination of RAG with embedding-based semantic similarity is methodologically well suited to low-resource settings: a pretrained embedding model carries most of the linguistic representation, a small curated reference corpus replaces the large labeled training set, and externally retrieved signals (search indexing, allowlists, public phishing databases) supply evidence that no static training set can reproduce. Lithuanian is therefore positioned not as an incidental regional case but as a deliberate test of an approach intended for languages where large labeled phishing corpora are unlikely to become available, and the same architecture is, in principle, transferable to other low-resource languages by replacing the reference corpus and the curated lists.

Drawing on Basil Bernstein's (1996) theory of pedagogic discourse, particularly the Pedagogic Recontextualizing Field (PRF), this study conceptualizes how a smishing risk detection methodology can be transformed into a potential learning resource for cybersecurity literacy. Therefore, developing an explainable smishing detection tool addresses both a technical and a cybersecurity education gap. By integrating RAG with transparent, criterion-based risk scoring, such a system can simultaneously identify threats and provide informative feedback to users - helping them understand why a message is risky and how to evaluate similar threats in the future. This dual functionality supports the development of critical digital literacy, a competency increasingly recognized as essential in modern education (Buchan et al., 2024). Despite advances in detection technologies, most systems remain black boxes, offering little transparency or educational value. This is a significant limitation in educational contexts, where promoting critical thinking, digital literacy, cybersecurity education, and trust in intelligent tools are essential. In this paper, we focus on the developing and initially evaluating the proposed tool to ensure its technical robustness and the potential for pedagogical reconceptualization and interpretability before deploying it for educational purposes.

While the proposed prototype is designed with the potential for pedagogical recontextualization to support cybersecurity literacy in educational settings, the primary aim of the present study is the technical development and evaluation of an explainable RAG-based smishing detection system for Lithuanian-language SMS messages. Although the prototype is designed for potential educational applicability, deployment in instructional settings requires technical reliability, interpretability, and cross-model consistency. Therefore, before examining pedagogical impact, it is necessary to establish whether the system produces stable, explainable, and reproducible outputs.

To address the identified technical and contextual gaps, particularly in a Lithuanian low-resource linguistic setting that motivates the use of retrieval-based approaches and semantic similarity instead of large supervised corpora, this study pursues the following research objectives:

1. To evaluate the cross-model consistency of phishing risk scores across different large language models.
2. To compare the clarity and reliability of phishing risk explanations generated by the tested models.
3. To determine the contribution of specific risk indicators to overall risk scores and system interpretability.
4. To assess the prototype's suitability for pedagogical reconceptualization as a foundation for future cybersecurity education use.

The following research questions address these foundational properties, which constitute prerequisites for educational integration.:

RQ1: How consistent are phishing risk scores across different large language models (GPT o1, Claude Sonnet 3.7, DeepSeek V3) when analyzing SMS messages?
RQ2: What differences exist among the tested LLMs in interpreting phishing risk factors reliably?
RQ3: Which specific risk indicators (e.g., semantic similarity, URL indexing, allowlist verification) contribute most significantly to the risk scores, and how do they influence the interpretability of the results?
RQ4: Does the developed prototype demonstrate structural and functional characteristics that make it suitable for pedagogical reconceptualization as an instructional tool in cybersecurity education?

Although the system architecture incorporates features that may support instructional use, such as transparent weighting of risk criteria and structured natural-language explanations, this study has not empirically evaluated the prototype's educational impact. Rather, the system is conceptualized as a foundational framework that future research could examine for its effectiveness in enhancing cybersecurity awareness, digital literacy, and critical evaluation skills in educational contexts. In sum, this study evaluates the stability and consistency of LLM-based risk scoring in an explainable RAG smishing detection prototype, thereby establishing a technical foundation for its future pedagogical recontextualization in cybersecurity education.

2. Literature review

2.1. Smishing trends, tactics, and detection approaches

The prevalence of smishing attacks has escalated in recent years, and cybercriminals have employed increasingly sophisticated tactics. For instance, the "Smishing Triad," a Chinese-speaking cybercriminal group, has been identified as a major actor in global smishing campaigns, utilizing advanced phishing kits and impersonating reputable institutions to deceive victims across more than 121 countries (Burgess, 2025). These attacks often involve messages that create a sense of urgency, prompting recipients to click on malicious links or provide personal information under the guise of resolving fictitious issues such as unpaid tolls or undelivered packages. Rahman et al. (2023) found that 16.92% of participants may have been victims of the initial smishing attack. Upon exposure to a second, differently worded smishing message,

12.82% of the randomly selected participants fell for the attack again.

To combat smishing, various detection mechanisms have been developed, primarily categorized into rule-based systems and machine learning (ML) classifiers.

Rule-based systems operate on predefined heuristics and patterns to identify malicious messages. For example, Jain and Gupta (2018) proposed a rule-based framework that analyzes SMS content for specific keywords and patterns that indicate smishing attempts. While these systems are easy to implement and understand, they often struggle to adapt to evolving cybercriminal tactics, resulting in decreased effectiveness over time. ML approaches offer a more dynamic solution by learning from data to identify smishing messages. Mishra and Soni (2020) developed a content-based model that uses features extracted from SMS messages and their associated URLs to train classifiers that distinguish between legitimate and malicious messages. Several more studies were done to analyze smishing message detection systems (El Karhani et al., 2023; Jain et al., 2025; Jain & Gupta, 2018, 2019; Joo et al., 2017; Mahmud et al., 2024; Maqsood et al., 2023; Mishra & Soni, 2020, 2023; Seo et al., 2024; Srinivasarao & Sharaff, 2023). These studies have demonstrated promising results, with some achieving detection accuracies exceeding 92% and even 99%. Despite the advancements in ML-based detection, challenges persist. The dynamic nature of smishing tactics requires continuous model updates and retraining to remain effective. Additionally, reliance on labeled datasets for supervised learning poses limitations, as acquiring comprehensive, up-to-date datasets can be resource-intensive.

While both rule-based and ML-based detection systems have their merits, they also exhibit limitations. Rule-based systems lack adaptability, and ML models require substantial data and computational resources. Moreover, many existing systems operate as "black boxes," providing limited transparency into their decision-making processes, which can erode user trust and understanding (Bhagat et al., 2023).

To address these challenges, integrating XAI into smishing detection frameworks is gaining attention. Researchers have proposed developing and implementing explainable artificial intelligence methods to enhance transparency and foster a greater understanding of AI behavior (Taylor, 2024). XAI aims to make AI decision-making processes transparent and understandable to users, thereby enhancing trust and facilitating better user cybersecurity education to recognize and avoid smishing attacks (Melo et al., 2022). However, XAI can also leave systems vulnerable to adversary attacks (Capuano et al., 2022).

2.2. Phishing and smishing user cybersecurity education

Sheng and colleagues found that individuals with higher education and employment experience were significantly better at identifying phishing emails, indicating greater personal awareness of security issues (Sheng et al., 2010). These findings underscore the importance of prioritizing cybersecurity education in workplaces and educational institutions, emphasizing the need to update training programs to address evolving threats continually. This includes raising awareness about SMS phishing risks and effective detection strategies.

Researchers have explored a variety of educational interventions to improve users' phishing-detection skills. Approaches such as interactive games, simulations, and contextual "teachable moments" embedded in messaging platforms have shown promise in enhancing people's ability to recognize phishing attempts (Kumarasinghe et al., 2023). For example, organizations often conduct phishing email drills or provide training modules to help users practice identifying suspicious signs. Some organizations conduct in-person briefings led by the chief information security officer for newly hired employees. In contrast, others reinforce cybersecurity awareness by sending short video messages to staff members who have previously been targeted by cyberattacks. Among the most widely used and effective methods for increasing awareness are simulated phishing campaigns (Callejas et al., 2021). When well-designed, these methods can produce measurable improvements in awareness and retention of safe behavior. Unfortunately, many individuals remain vulnerable – especially as attack vectors have expanded from email to mobile SMS. Human error remains a leading cause of security breaches.

Edwards et al. (2023) found that while most college-aged participants recognized in theory that clicking links from unknown SMS senders is dangerous, in practice, a majority still clicked such links. Intriguingly, participants often misunderstood the concept of "SMiShing" and relied on superficial cues (e.g., the sender's number or typos) to judge a message's legitimacy rather than deeper indicators (e.g., unusual sending times or mass messaging). This behavior–attitude gap highlights the ongoing need to develop cybersecurity literacy, particularly smishing-specific awareness: users may generally be aware of "online scams" yet lack the knowledge or habits to apply that caution to text messaging contexts. Indeed, the human factors literature identifies insufficient user knowledge and critical thinking about SMS phishing as significant challenges. Thus, building user capacity to detect smishing requires targeted educational tools that not only warn about risks but also cultivate the analytical skills to scrutinize mobile messages. The present work addresses this gap by providing a smishing detection prototype that supports both security and educational objectives: it aims to strengthen users' critical evaluation of SMS content by making the detection process transparent and informative.

2.3. Digital literacy and cybersecurity education in schools

Broader digital literacy and cybersecurity education initiatives provide important context for phishing and smishing training, especially in formal educational settings. Digital literacy is now recognized as a critical life skill for youth and adults to navigate an information-rich, networked world safely (Buchan et al., 2024). Low digital literacy has been linked to a higher susceptibility to online threats, prompting calls for dedicated programs to teach safe and responsible technology use (Buchan et al., 2024; Ferrari et al., 2024). In response, governments and organizations (e.g., UNESCO's Digital Literacy Global Framework) have begun promoting curricula that include topics such as privacy, online safety, and cyber hygiene as part of quality education goals (Buchan et al., 2024; Crabb et al., 2024).

However, integrating cybersecurity topics into classrooms comes with challenges. Many resources have been developed for teaching cybersecurity to children and adolescents (e.g., videos, games, lesson plans), but evidence of their effectiveness is limited. A systematic review by Zhang-Kennedy and Chiasson (2021) identified dozens of multimedia cybersecurity tools designed for youth and found that approximately 43% targeted children or teens; however, most lacked evaluations to support their effectiveness in achieving broader learning outcomes (McLeod et al., 2024). Teachers play a pivotal role in cybersecurity education, but they often lack the training and support needed to teach these topics confidently. Teachers often address cybersecurity and privacy from a safety-focused, reactive stance, relying on informal methods such as class discussions rather than formal assessments. While content creators understand students' needs, they often fail to tailor resources to support teachers effectively. As a result, much of the available cybersecurity material, especially games, remains underutilized in classrooms due to its inflexible format (Pramod, 2025). Teachers prefer adaptable tools, such as lesson plans and short videos (Stojšić, 2024). Aligning resource design with teachers' needs could lead to broader classroom use and improved student understanding of cybersecurity and privacy (McLeod et al., 2024; Sales et al., 2024). In one study, Italian educators acknowledged that their students should be better prepared to recognize online risks but also admitted that they needed additional training to enhance their awareness and ability to teach these topics (Corradini & Nardelli, 2020). Unfortunately, formal training resources for teachers remain scarce; existing programs tend to focus narrowly on issues such as student data privacy compliance, with few comprehensive cybersecurity training courses available for educators (McLeod et al., 2024).

This misalignment is also evident in the design of educational materials: curriculum developers ("creators") often produce resources based on the latest threats or funding priorities, without centralized guidelines for core cybersecurity topics (McLeod et al., 2024). The result is an abundance of ad hoc materials that teachers must sift through and adapt, often without knowing whether they meet pedagogical goals.

These findings emphasize that simply providing security tools or curricula is not enough; supporting educators is equally important. Effective adoption of classroom cybersecurity lessons requires clear guidelines, teacher-friendly tools, and evidence that the materials work as intended (McLeod et al., 2024). By focusing on explainability and educational feedback, our approach aligns with calls to better tailor cybersecurity tools to learning environments. An interpretable smishing detection system can also serve as a teaching aid, enabling educators (even non-experts) to confidently demonstrate how to analyze suspicious messages and foster classroom discussions on why specific messages are dangerous.

Bernstein's (1996) theory of pedagogic discourse, particularly the concept of the Pedagogic Recontextualizing Field (PRF), provides a theoretical basis for understanding how professional cybersecurity knowledge can be reorganized as educational content. Within this framework, teachers, educational institutions, and curriculum developers may selectively appropriate and transform knowledge produced outside formal education. In the present study, the smishing risk detection methodology is conceptualized as a possible object for such recontextualization: its criteria, decision procedures, and explanatory outputs may be reorganized into learning materials that support cybersecurity literacy. This framing positions the prototype not as an empirically validated educational intervention, but as a technical foundation that may later be adapted for instructional use. In essence, the system is designed to bridge the gap between abstract cybersecurity concepts and concrete, observable examples that students and teachers can explore together, thereby integrating digital literacy objectives into schools and daily routines.

2.4. Interpretability of AI systems in educational tools

XAI has been introduced as a promising approach to enhance the transparency of AI systems, thereby facilitating their broader adoption in high-stakes or sensitive domains (Barredo Arrieta et al., 2020). It primarily aims to develop methods that enable end-users to understand, trust, and interact more effectively with modern AI technologies. One emerging solution is to integrate XAI techniques, enabling models not only to predict threats but also to provide human-understandable justifications. XAI aims to transform the AI into a "grey box" that reveals its decision logic without sacrificing much accuracy (Melo et al., 2022). This transparency is particularly critical in an educational context: it enables learners (or instructors) to see the rationale behind AI judgments, fostering deeper understanding and critical analysis rather than blind acceptance, and supporting the development of cybersecurity literacy.

Recent studies have begun to demonstrate the benefits of XAI for both learners and tool developers. XAI serves as a "facilitator for educational professionals" by mitigating biases and supporting ethical, auditable use of AI on school data, ultimately producing models that are "more adequate to a school context" (Melo et al., 2022; Simbeck, 2024). In other words, when teachers and administrators understand AI's reasoning, they are more likely to trust and use its insights effectively in decision-making, thereby contributing to the development of cybersecurity literacy (Kim, 2024; Melo et al., 2022). This aligns with broader ethical guidelines that view transparency as key to responsible AI deployment in education. An interpretable model respects user autonomy and accountability, whereas an inscrutable model might inadvertently reinforce biases or deprive learners of the opportunity to question and learn from the AI's output (Melo et al., 2022). Indeed, Melo et al. note that incorporating XAI is part of ensuring an "ethical, transparent, and auditable" use of educational data (Melo et al., 2022), echoing the sentiment that students and educators should be able to audit and challenge AI decisions (Simbeck, 2024), thereby supporting the development of cybersecurity literacy.

In cybersecurity, explainable AI is proving equally valuable. Lim et al. (2025) introduce EXPLICATE, an explainable phishing detection framework, to tackle the "critical divide" between automated AI judgments and user trust (Lim et al., 2025). Their system combines a high-accuracy phishing classifier with dual explanation layers - LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) - and an LLM-based module to translate technical features into natural-language explanations (Lim et al., 2025). The result is a tool that achieved state-of-the-art 98% detection accuracy while providing reasoned explanations, giving users insight into why an email was classified as malicious (Lim et al., 2025). Notably, Lim et al. report that such "high detection performance can go hand-in-hand with meaningful explainability", addressing the long-held concern that adding transparency might come at the cost of accuracy (Lim et al., 2025). By evaluating explanation quality (measured as the consistency between the model's features and the LLM-generated narrative, which was 96.8% in their case), they demonstrate that it is feasible to generate explanations that are both faithful to the model and accessible to users (Lim et al., 2025). This approach targets user education, as the explanations help demystify phishing cues (e.g., why a specific URL or sender raises suspicion) and thereby teach users how to spot scams in the future. The positive results from EXPLICATE underscore a key point for our study: transparency can significantly improve user trust and understanding without degrading AI's protective (Lim et al., 2025). In an educational setting, this means students can learn from AI (by observing its reasoning) rather than just receiving warnings (Darwin et al., 2024), thereby enabling AI to serve as an instructional tool for developing cybersecurity literacy.

Our work builds on these insights by employing a RAG approach for explainable smishing detection. RAG is an emerging technique in which an AI model retrieves relevant knowledge (e.g., known scam URL databases, search engine results for a domain) and uses that information to inform its output generation. By design, RAG enables the use of external or up-to-date information by retrieving relevant content at the time of the request to inform the AI's response, thereby providing users with context that can help them understand the factual basis for risk assessment. This differs from post hoc explanation methods, such as LIME, by integrating interpretability into the model's operation. A literature search was conducted in the Scopus database using the keywords "smishing OR phishing; Retrieval-Augmented Generation" to understand what research has already been done. The relevant literature is primarily found in specialized security and data mining conferences, such as the ACM International Workshop on Security and Privacy Analytics (IWSPA), the ACM Conference on Data and Application Security and Privacy (CODASPY), and the IEEE Conference on Dependable and Secure Computing concentrated between 2024 and early 2025 (Boumber et al., 2024; Chen et al., 2025; Ferrag et al., 2025; Lee et al., 2024; Manor et al., 2024; Park et al., 2025; Zaw et al., 2025). Based on the these articles, we can conclude that no prior peer-reviewed work has applied RAG to phishing or smishing detection in an educational context, making our approach exploratory. Nonetheless, it aligns with trends in XAI, as seen in Lim et al.'s (2025) work, which prioritizes clear, user-facing rationale. In our system, a suspicious SMS is analyzed by checking, for example, whether the embedded URL appears on the open web and whether the message matches known malicious patterns. These checks are then explicitly reported to the user as part of the AI's reasoning. By assigning weights to features (e.g., "URL is not indexed by search engine" might contribute a certain percentage to the risk score), the tool offers a transparent breakdown that the user (or a teacher using the tool in class) can follow. This design addresses the cybersecurity educational and ethical priorities identified in the literature, seeking to empower users with understanding rather than just issuing warnings. In

terms of educational value, such transparency encourages critical thinking – users learn why something is unsafe, thereby reinforcing their phishing literacy. Ethically, it promotes accountability, as the AI's decision-making process is open to inspection, thereby reducing the likelihood of unchecked biases or errors.

The literature (Lim et al., 2025; Melo et al., 2022; Simbeck, 2024) supports our study's dual focus on detection and education in the integration of explainable AI to foster trust, enhance understanding, and promote the development of cybersecurity literacy. Our prototype differentiates itself by using a RAG-based solution to deliver an explainable smishing risk assessment, directly addressing gaps noted in prior work – namely, the lack of attention to smishing threats in user education and the obscure nature of other security tools. By comparing the consistency of its risk evaluations, we also respond to calls for rigorous evaluation of educational cybersecurity tools (McLeod et al., 2024). If future studies find this approach technically reliable and pedagogically validated, it may inform the development of AI-supported educational tools that help learners understand how smishing risks are assessed. This aligns with the broader goal in the literature of fostering a culture of digital literacy and critical thinking in cybersecurity, ultimately empowering users to be less susceptible to phishing and smishing attacks.

3. Methodology

3.1. System architecture

To enhance usability, the proposed prototype initiates its analysis by taking a screenshot of the user's SMS message. Essential data elements, including the sender's information, message content, timestamp, and hyperlinks, are extracted from this screenshot for further evaluation (see Fig. 1).

Hyperlinks identified within the message are subjected to a domain verification procedure against search engine indexing. Domains not indexed by search engines pose a heightened phishing risk, as they may be newly created for malicious intent.

Subsequently, domains are cross-referenced against an internal allowlist database of recognized websites from prominent organizations, allowing immediate exclusion of known legitimate sources often used in

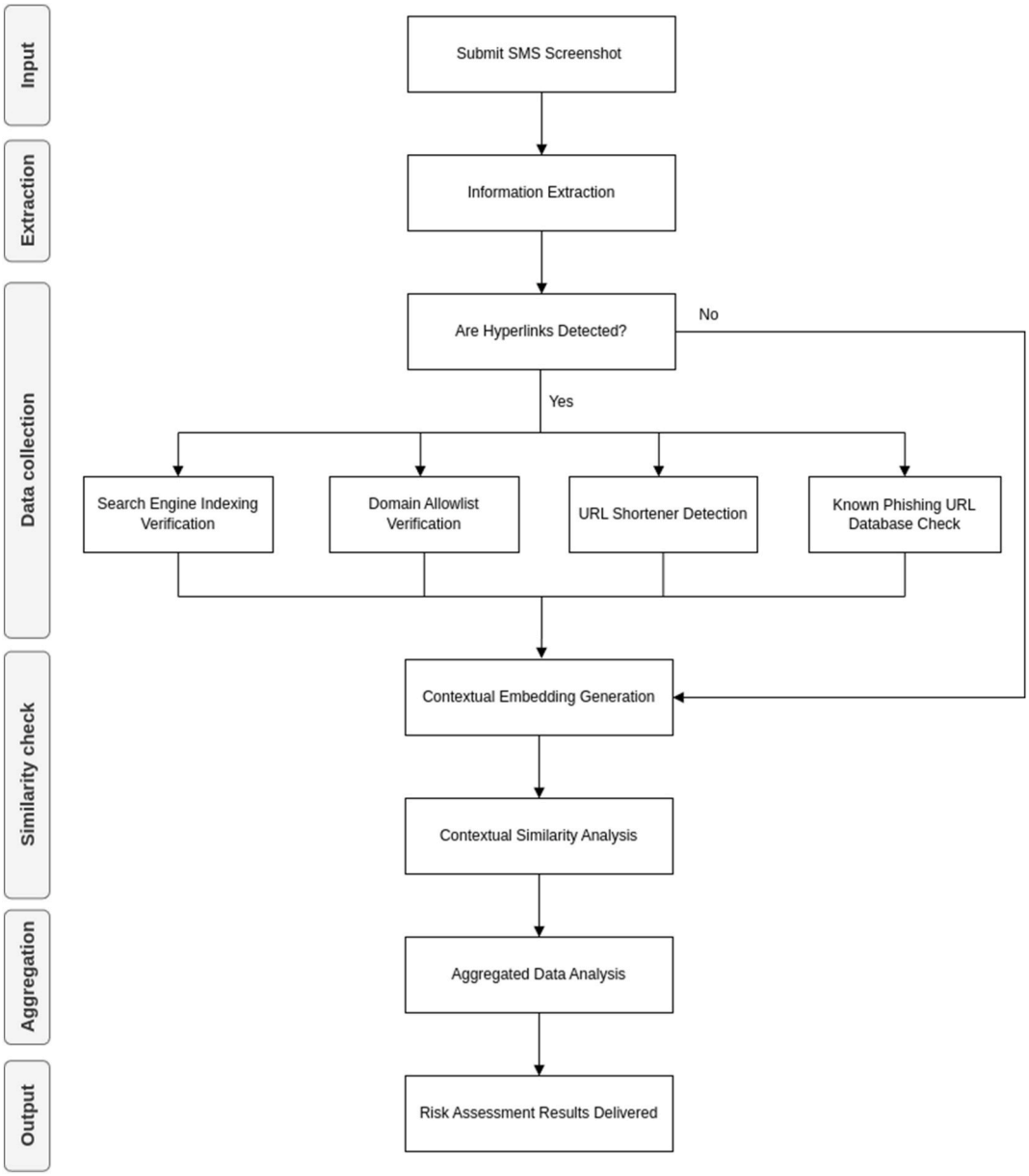


Fig. 1. System architecture for RAG-Based smishing risk assessment workflow.

phishing attacks from suspicion of phishing.

Additionally, domains are checked against an internal database of URL shortening services; hyperlinks using these services are flagged as having a higher risk due to their prevalence in phishing schemes.

In the next analytical step, the message content is transformed into text embeddings, enabling a semantic comparison with an internal database of previously assessed messages. This comparison aims to identify contextual and linguistic similarities to known phishing or benign messages.

All OpenAI models (GPT-4o version 2024-08-06, text-embedding-3-large, and o1 version 2024-12-17) were deployed via Azure OpenAI Service. DeepSeek V3 was deployed through the Azure AI Foundry model catalog, and Claude Sonnet 3.7 (claude-3-7-sonnet-20250219) was accessed via the Anthropic API.

Finally, the collected data is passed to a large language model (LLM), configured to reduce output randomness to the extent permitted by each provider's API: Claude Sonnet 3.7 (Anthropic API, model ID claude-3-7-sonnet-20250219) and DeepSeek V3 (Azure AI Foundry, model ID DeepSeek-V3-0324) were called with temperature = 0 and top_p = 1.0, whereas o1 (Azure OpenAI Service, model version 2024-12-17) does not expose temperature or top_p parameters and was therefore called with default settings, relying on its internal reasoning process. Because randomness-related parameters cannot be controlled uniformly across the three models, full determinism is not assumed for any of them. The LLM evaluates each message against five predefined criteria: (1) search engine indexing status, (2) domain allowlist verification, (3) URL shortener detection, (4) presence in known phishing databases, and (5) semantic similarity to known phishing or benign messages. Each criterion is assessed individually. The LLM receives the binary or categorical outcome of each check (e.g., indexed/non-indexed, allowlisted/not allowlisted) together with the numeric cosine-similarity score, assigns dynamic integer weights (constrained to sum to exactly 100%) reflecting each criterion's contextual relevance, and produces a single aggregated risk score on a 0-100 scale (full prompt text in Appendix A). The prompt does not prescribe specific weight values or priority orderings for any criterion; it only requires consistency and relevance to phishing risk, so the observed allocations (Table 2) reflect each model's independent judgment. The LLM generates a structured JavaScript Object Notation (JSON) output containing the assigned weights, the overall risk score, and a detailed explanation.

Each criterion is assessed individually, and dynamic integer weights (summing to 100%) are assigned to reflect each factor's relative importance in the current context. The LLM then computes a weighted risk score on a scale from 0 to 100. It generates a structured JavaScript Object Notation (JSON) output containing both the score and a detailed explanation, delivered in the specified language, describing how each criterion contributed to the final risk classification.

In the present study, RQ4 is operationalized through technical indicators of pedagogical reconceptualization developed to evaluate materials for instructional suitability, including explanation transparency, criterion-level traceability, and scoring consistency. Direct educational impact (e.g., learning gains or behavioral change) is outside the scope of this investigation; however, the study draws on Bernstein's (1996) concept of pedagogical recontextualization to argue that the smishing detection methodology can be applied in educational contexts to support the development of cybersecurity literacy, thereby demonstrating its educational value.

3.2. Information extraction

The first stage involves collecting SMS messages from public sources (e.g., Social Networks) and extracting SMS screenshot details (message text, sender, hyperlinks, timestamp) from image form into analyzable text. The authors have manually reviewed all evaluation screenshots and reference messages shared in the project repository to redact personally identifiable information, such as recipient phone numbers, document IDs, parcel links, and so on. We used the multimodal capabilities of GPT-4o (Azure OpenAI Service, model version 2024-11-20) to perform optical character recognition (OCR) on SMS message screenshots. This step is crucial as it converts raw visual data into readable text for subsequent analysis (Shinde et al., 2024). Recent research emphasizes the importance of this phase. Shinde et al. (2024) demonstrate that applying OCR (e.g., Tesseract) to image-based SMS spam provides a foundation for accurate classification. By using the GPT-4o API for OCR, our system achieves higher accuracy than standard OCR models (e.g., Tesseract, EasyOCR) and enables simpler user data collection.

3.3. Link and domain analysis

3.3.1. Search engine indexing verification

All URLs found in the message are checked to see whether their domains appear in public search engine results. For each URL, the host is extracted and a site:{domain} query is issued to a Serper.dev Google search API requesting a single result. If at least one result is returned, the domain is classified as indexed; an empty result set classifies it as not indexed. No locale or language parameters are set. The project repository includes a Python script (search-index-check.py) that reproduces this check. The absence of a domain from search results is treated as a phishing risk indicator on the premise that legitimate websites are usually indexed.

In contrast, malicious domains are often too new or short-lived to be indexed (Cao et al., 2025). In our implementation, a non-indexed domain does not necessarily indicate a malicious site. However, it raises a red flag for that URL, assuming that benign domains of established organizations should be visible online.

3.3.2. Domain allowlist verification

To reduce false positive results, each extracted URL's domain is also cross-referenced against a manually curated internal allowlist of 30 known trusted Lithuanian domains, covering major banks (e.g., Swedbank, SEB, Luminor), government institutions (e.g., VMI, SoDra, Regitra), postal and courier services (e.g., Lietuvos paštas, Omniva, DPD, Venipak), telecommunications providers (e.g., Tele2, Telia), and other popular local services. The authors compiled the list based on domains commonly impersonated in Lithuanian smishing campaigns, and it is publicly available in the project repository (https://github.com/LU-IZPF/smishing). Because the allowlist is small and context-specific, it is maintained and extended manually as new impersonation targets emerge; no automated update cadence is currently applied. If the domain is found on this allowlist, the information is used to downgrade the phishing risk associated with that message. Attackers often register look-alike domains or impersonate well-known organizations; therefore, an exact match with a legitimate domain is likely benign or, at the very least, not a new phishing site. Recent studies on phishing defense have highlighted that maintaining an allowlist of verified domains can significantly reduce the flagging of legitimate sites (Bayer et al., 2023). In our system's workflow, if a URL's domain appears on the allowlist, subsequent risk checks will still be executed; however, the final risk score for that message will reflect a lower likelihood of fraud. This mechanism aligns with best practices in security filters, which combine blocklists with allowlists to balance sensitivity and precision (Bayer et al., 2023).

3.3.3. URL shortener detection

Cybercriminals often use URL shortening services to hide the true destination of phishing links (Le Page et al., 2018). To address this, our system maintains a curated list of 657 known URL shortener domains (e.g., bit.ly, tinyurl.com, goo.gl, is.gd). The list was aggregated from publicly available shortener-domain compilations and is provided in the project repository (https://github.com/LU-IZPF/smishing). Any hyperlink in the SMS that matches one of these shortening services is automatically flagged. The presence of a shortener can increase the message's

risk score because it introduces ambiguity about the final URL and is a known tactic in phishing and smishing campaigns (Zhang et al., 2025). Recent security research confirms that generic shorteners, such as bit.ly, are "notorious for being abused in cyber crimes" (Zhang et al., 2025) and are often used to lure users by masking malicious domains. In the context of SMS phishing, scammers leverage short URLs to both conserve character space and make malicious links less obvious on small mobile screens.

3.3.4. Known phishing URL database check

In this step, any URLs found in the SMS message are extracted and checked against PhishTank (https://phishtank.org), a public repository of phishing URLs continuously updated through crowdsourced submissions. If the URL extracted from the SMS is found in such a database, this finding is retained for subsequent analysis to determine the final risk score. If no match is found, this does not guarantee safety; rather, it indicates that the URL is not listed in public phishing databases at the time of the query.

The PhishTank database check is a powerful way to catch known phishing attacks and minimize false positives. The approach offers near-certain identification when a match occurs, leveraging the collective intelligence of community-submitted threat reports. However, URL blocklists alone cannot effectively detect novel or rapidly evolving phishing attacks (Ghalechyan et al., 2024). In this prototype, PhishTank was used for URL verification, although other public databases, such as Google Safe Browsing and URLhaus, offer similar functionality.

3.4. Contextual similarity analysis

Beyond surface-level URL and message text checks, our system evaluates the message's textual content for similarity to known phishing and benign messages. We generate a semantic embedding of the SMS text using OpenAI's transformer-based embedding model text-embedding-3-large (Azure OpenAI Service). This model converts the message into a numerical vector representation in a 3072-dimensional semantic space, capturing nuanced linguistic and contextual features (Alhuzali et al., 2025). The use of transformer-derived embeddings is motivated by recent advances in phishing detection: transformer encoders like Bidirectional Encoder Representations from Transformers (BERT) and its variants can automatically learn the subtle cues of phishing language (e.g., urgent tone, request for credentials, suspicious wording) far better than manual feature engineering (Alhuzali et al., 2025; Chrysanthou et al., 2024). These embeddings enable the system to move beyond keyword matching and instead assess the message's "meaning", similar scenarios, and writing style in comparison to a reference set of messages. We maintain a reference dataset of 78 Lithuanian-language SMS messages (67 labeled as phishing, 11 as benign) whose embeddings serve as the comparison corpus. The phishing examples were collected from publicly shared Lithuanian smishing reports on social media and news sources, as well as from direct contributions by individuals who received such messages; the benign examples were drawn from authentic, non-malicious SMS messages. The authors manually reviewed and labeled each message. The dataset is publicly available in the project repository (https://github.com/LU-IZPF/smishing). The text of every reference message was embedded using OpenAI's text-embedding-3-large model, producing three 072-dimensional vectors that were stored and reused across evaluations. For each new message, its embedding is compared to those in the reference set using cosine similarity. A configurable cosine similarity threshold is applied to identify reference messages that are sufficiently close to the input; only those exceeding the threshold are surfaced to the LLM as contextual evidence alongside the raw similarity score. The threshold value was selected empirically during development by observing retrieval quality on the current reference corpus; the full set of labeled messages with their precomputed embeddings is published in the project repository, enabling independent verification and threshold tuning. Importantly, the threshold serves solely as a retrieval gate; it does not determine the final phishing classification, which the LLM makes by interpreting the similarity score together with the other four criteria to produce the continuous 0–100 risk score. The threshold value can be adjusted as the reference database grows or as new phishing writing styles emerge.

3.5. LLM prompt and aggregation

The prototype intentionally presents the raw 0–100 risk score rather than mapping it to discrete risk categories (e.g., low/medium/high). As Table 1 illustrates, score distributions for phishing and benign messages overlap across models. For example, DeepSeek V3 assigned scores as low as 31 to confirmed phishing messages (F10) and as high as 53.5 to benign messages (B9), making fixed thresholds unreliable for categorical classification. Instead, the continuous score, together with the per-criterion explanation, is designed to prompt the user's judgment, aligning with the prototype's educational objective of fostering critical evaluation rather than binary accept/reject decisions.

After completing every prior stage - OCR extraction, search-engine-indexing check, domain-allowlist match, URL-shortener flag, phishing-database lookup, and semantic-similarity scoring – the resulting values are assembled into a compact record and handed to the LLM. The prompt is split into two parts: first, the main LLM analysis instructions and output format; second, the data collected from prior stages. Guided by the schema (full text in Appendix A), the model calculates a weighted phishing risk score and returns it, along with a criterion-by-criterion rationale, risk score, and feedback. LLM's judgment remains fully traceable to concrete facts already established by the system. Because the schema fixes the set of criteria, the integer-weight constraint (sum to 100%), and the structure of the output, the same JSON record serves both as a machine-parseable artifact for independent auditing and reproducibility checks and as a stable instructional template that can be reused across messages and across teaching sessions without altering the underlying detection logic. Three language models developed by three independent providers were selected to compare risk score generation across distinct model families while holding the surrounding pipeline constant: o1 (OpenAI), Claude Sonnet 3.7 (Anthropic), and DeepSeek V3. This cross-provider design was intended to reduce the risk that observed stability and weighting patterns reflect characteristics of a single model family. Because LLM outputs are not fully deterministic even at minimal temperature settings, each message was evaluated multiple times per model to capture output variability and compute meaningful descriptive statistics (mean, standard deviation, coefficient of variation) for each message–model pair. The number of repetitions was set to 10 per message–model pair (600 total evaluations across 20 messages and 3 models), which was sufficient to reveal systematic patterns in score variability across models while remaining practically feasible in terms of API cost. This constitutes an exploratory evaluation; a larger sample of repetitions would be needed to support formal inferential claims.

4. Evaluation and results

To evaluate the consistency and variance of LLM-based phishing risk assessments, 20 SMS messages were analyzed. Each message was evaluated 10 times independently by three language models developed by three independent providers to ensure that observations would not be confined to a single model family. All models used the same prompt structure and data inputs and were required to return a structured risk report with per-criterion weighting and final scores, so that any differences in output reflect model behavior rather than pipeline variation. For each message-model pair, Table 1 reports the mean risk score averaged over the 10 repetitions, the standard deviation, and the coefficient of variation (CV).

Table 1
Risk scores.

Risk Scores[a]									
	GPT o1			DeepSeek V3			Claude Sonnet 3.7		
MSG ID	Avg. Mean	Std. dev.[b]	CV	Avg. Mean	Std. dev.	CV	Avg. Mean	Std. dev.	CV
F1	75.2	6.8819	0.0915	60	4.669	0.0778	85	0	0
F2	85.4	3.6387	0.0426	71.7	3.2879	0.0459	83.2	4.7497	0.0571
F3	61.3	5.2735	0.086	43.4	6.2322	0.1436	70.7	3.9762	0.0562
F4	77	6.0828	0.079	66.5	3.3838	0.0509	78.6	1.2806	0.0163
F5	66	5.2536	0.0796	56.5	5.5	0.0973	74.6	2.7641	0.0371
F6	82	4.1231	0.0503	62.6	3.6661	0.0586	81.2	5.3442	0.0658
F7	78	6.3403	0.0813	64.8	3.0265	0.0467	87	4.5826	0.0527
F8	70.5	9.8615	0.1399	60	5.9161	0.0986	85	0	0
F9	55.2	7.8333	0.1419	34	2	0.0588	78	4.5826	0.0588
F10	52	9.2736	0.1783	31	3	0.0968	71.5	7.7621	0.1086
B1	9.8	3.9192	0.3999	10	0	0	5.5	1.5	0.2727
B2	4.4	1.8547	0.4215	9	2	0.2222	5.5	1.5	0.2727
B3	32.2	6.8088	0.2115	18.4	4.9031	0.2665	23	3.3166	0.1442
B4	11	3.2558	0.296	10	0	0	11	2	0.1818
B5	60.7	6.8855	0.1134	11	11.3578	1.0325	64	11.3578	0.1775
B6	33.7	14.9937	0.4449	25	4.4721	0.1789	23	2.4495	0.1065
B7	34.5	7.8899	0.2287	25	5	0.2	15.5	1.5	0.0968
B8	60.6	8.015	0.1323	24	14.9666	0.6236	55.5	2.6926	0.0485
B9	65.5	10.1119	0.1544	53.5	3.9051	0.073	65.5	3.5	0.0534
B10	16.5	3.8013	0.2304	10	0	0	14	2	0.1429
Avg.	-	**6.6049**	**0.1802**	-	**4.3643**	**0.1686**	-	**3.3429**	**0.0975**

[a] For each message-model pair, the 95% confidence interval can be computed as Mean $\pm$ 2.262 $\times$ SD/$\sqrt{10}$ (t-distribution, df = 9).
[b] Avg. Mean is the mean of 10 runs.

4.1. Risk score stability

Table 1 summarizes the mean risk score, standard deviation, and coefficient of variation (CV) for each message and model. The CV was used to assess relative score volatility, with the following pre-defined thresholds adopted for interpretation: CV $\leq$ 0.10 indicates high stability (scores vary by no more than 10% of the mean); 0.10 < CV $\leq$ 0.20 indicates moderate stability; 0.20 < CV $\leq$ 0.30 indicates low stability; and CV > 0.30 indicates an unstable output.

Observations:

- Claude Sonnet 3.7 displayed the lowest overall CV across all messages (avg. CV = 0.0975), falling within the high-stability band (CV $\leq$ 0.10) and indicating strong internal consistency.
- DeepSeek V3 (avg. CV = 0.1686) and GPT o1 (avg. CV = 0.1802) both fall within the moderate-stability band (0.10 < CV $\leq$ 0.20). However, GPT o1 produced several per-message CVs in the low-stability (e.g., B3: 0.21; B7: 0.23; B4: 0.30) and unstable ranges (B1: 0.40; B2: 0.42; B6: 0.44), indicating notably inconsistent scoring on certain benign messages. Appendix B reports the minimum, quartiles, median, and maximum scores for these high-variability messages.
- To complement the CV analysis, the two-way mixed-effects, single-measures, consistency intra-class correlation coefficient ICC (3,1) (Koo & Li, 2016) was computed for each model across the 10 repetitions. Claude Sonnet 3.7 obtained an ICC of 0.980 (95% CI [0.964, 0.991]), DeepSeek V3 obtained 0.935 (95% CI [0.888, 0.969]), and GPT o1 obtained 0.920 (95% CI [0.862, 0.962]). All three values exceed the 0.90 threshold for excellent reliability (Koo & Li, 2016), confirming that the repeated evaluations produce highly consistent score patterns across messages. The ordering mirrors the CV-based findings: Claude Sonnet 3.7 is the most reliable, followed by DeepSeek V3 and GPT o1.
- A Friedman test, with the message as the unit of analysis (N = 20) and the per-message CV as the dependent variable, confirmed that CVs differ significantly across the three models, $\chi^2(2) = 8.84$, p = .012, Kendall's W = 0.22. Post-hoc Wilcoxon signed-rank tests with Bonferroni correction (paired by message) showed that Claude Sonnet 3.7 produced significantly lower CVs than GPT o1 (p < .001), while the differences between Claude Sonnet 3.7 and DeepSeek V3 (p = .724) and between GPT o1 and DeepSeek V3 (p = .429) were not statistically significant.
- In phishing messages (F1-F10), all three models generally assigned higher average scores, consistent with expected threat levels. The aggregate phishing means were 79.48 (95% CI [75.34, 83.62]) for Claude Sonnet 3.7, 70.26 (95% CI [62.15, 78.37]) for GPT o1, and 55.05 (95% CI [45.02, 65.08]) for DeepSeek V3. The Claude Sonnet 3.7 and DeepSeek V3 intervals do not overlap, indicating a statistically meaningful difference in phishing scoring, while GPT o1 overlaps with both.
- For benign messages (B1-B10), the aggregate means were 32.89 (95% CI [16.54, 49.24]) for GPT o1, 28.25 (95% CI [11.12, 45.38]) for Claude Sonnet 3.7, and 19.59 (95% CI [9.78, 29.40]) for DeepSeek V3. All three intervals overlap substantially, suggesting no significant difference in benign-message scoring. GPT o1 tended toward higher and more variable scores (e.g., B5, B6, B8).

This variance suggests that while all models identify general risk trends, GPT o1 may be more sensitive or prone to overfitting certain features. At the same time, Claude Sonnet 3.7 tends toward conservative, reproducible outputs.

4.2. Weight distribution consistency

Table 2 shows the average weights (in %) assigned by LLMs to each evaluation criterion across all messages.

Observations:

- Semantic similarity received the highest cross-model average weight (23.5%) and was the top-ranked criterion for GPT o1 (27.4%). DeepSeek V3 and Claude Sonnet 3.7 ranked it second, marginally below domain allowlist verification (25% vs. 20.6% and 23% vs. 22.4%, respectively).
- DeepSeek V3 assigns slightly more weight to search indexing and allowlist verification, which may explain its tendency to give lower scores for benign messages with indexed, allowlisted domains.

Table 2
Weights.

Weights			
Criterion	GPT o1 Avg. Weight (%)	DeepSeek V3 Avg. Weight (%)	Claude Sonnet 3.7 Avg. Weight (%)
Search Engine Indexing Status	17%	22%	19.7%
Domain Allowlist Verification	21%	25%	23%
URL Shortener Detection	15%	15%	15.3%
Known Phishing Database Check	18.9%	18%	20%
Semantic Similarity Scores	27.4%	20.6%	22.4%

- All models treat URL shortening detection as the least influential factor (approximately 15%), though it remains consistently included.
- While all three models agreed on URL shortener detection as the least influential factor (~15%), they diverged on the relative importance of the remaining criteria - most notably semantic similarity, which GPT o1 weighted higher (27.4%) than DeepSeek V3 (20.6%) or Claude Sonnet 3.7 (22.4%). These differences emerged without prescribed weight values in the prompt, reflecting each model's independent interpretation.

To illustrate the system's complete output, Fig. 2 presents the JSON response generated by Claude Sonnet 3.7 for a phishing message impersonating the Lithuanian e-services portal (epaslaugos.lt). The output comprises three components: a natural-language report explaining each criterion's contribution, dynamically assigned integer weights that sum to 100%, and the overall risk score. To see contrast in explanations and weights across classes, see Appendix C for a benign example of DeepSeek V3.


```
{
  "report": "Šis SMS pranešimas kelia didelį įtarimą dėl
    sukčiavimo. Analizuojant penkis pagrindinius kriterijus:
    1) Domeno indeksavimo statusas: Nuorodos domenas
    'epaslaugos.lt-zinutes-e-portal.net' nėra indeksuotas
    paieškos variklių, kas rodo aukštą riziką. Netikri
    domenai dažnai nebūna indeksuoti. 2) Domeno patikimų
    sąrašų patikrinimas: Domenas nerastas patikimuose
    sąrašuose, kas didina riziką. 3) URL trumpinimo
    aptikimas: URL trumpinimas neaptiktas, kas šiek tiek
    mažina riziką. 4) Žinomų sukčiavimo duomenų bazių
    patikrinimas: Tiesioginio atitikmens sukčiavimo duomenų
    bazėse nerasta, tačiau tai nereiškia, kad domenas
    saugus. 5) Semantinis panašumas: Labai aukštas
    panašumas (0.87) su žinomu sukčiavimu pranešimu apie
    VMI baudas. Abi žinutės naudoja panašų šabloną -
    nurodo į baudas ir reikalauja prisijungti prie netikro
    portalo.",
  "weights": {
    "Search_Engine_Indexing_Status": 25,
    "Domain_Whitelist_Verification": 25,
    "URL_Shortener_Detection": 10,
    "Known_Phishing_Database_Check": 15,
    "Semantic_Similarity_Scores": 25
  },
  "score": 85
}
```


Fig. 2. Example JSON output generated by Claude Sonnet 3.7 for a phishing SMS.

5. Discussion

Integrating RAG into smishing detection frameworks represents a meaningful innovation in cybersecurity. Unlike conventional detection methods such as rule-based heuristics or static supervised machine learning, RAG dynamically retrieves and utilizes the latest external knowledge during inference, enhancing both adaptability and interpretability, and these results are in line with conclusions from previous research (Capuano et al., 2022; Melo et al., 2022; Taylor, 2024). Our results support this advancement: across 20 SMS messages evaluated with three LLMs, the Claude Sonnet 3.7 model consistently exhibited the lowest coefficient of variation (CV ≈ 0.0975), indicating strong stability and reproducibility - a crucial attribute for educational deployment.

Notably, the RAG-based design enables semantic similarity analysis and database cross-referencing to converge, potentially enhancing detection robustness against novel phishing tactics; however, the prototype has not been stress-tested against deliberate adversarial evasion techniques.

A critical barrier to effective phishing mitigation is not merely the lack of detection but the gap in user understanding and digital literacy. While conventional machine learning tools label messages as "phishing" or "safe," they rarely provide reasoning in a format accessible to non-experts. This reinforces users' reliance on opaque systems without increasing their resilience (Lim et al., 2025).

While many traditional phishing detection approaches - particularly rule-based systems - are architecturally interpretable in that their logic can be traced by developers through explicit heuristics or decision rules (Jain & Gupta, 2018), such transparency does not necessarily extend to end users. A rule-based system may be fully understandable at the implementation level yet still function as a practical "black box" from a layperson's perspective if it does not communicate its reasoning in accessible terms. The distinction is therefore not between interpretable and non-interpretable systems at the architectural level, but between systems interpretable to developers and explainable to users.

The proposed RAG-based framework contributes user-facing explainability rather than merely structural transparency and, beyond explainability, a reusable criterion-based structure. Instead of exposing raw features or triggered rules, the system produces structured natural-language explanations that explicitly describe how each criterion, such as domain indexing status, allowlist verification, or semantic similarity, contributed to the final risk score. By presenting dynamically assigned weights alongside narrative reasoning, the system helps users understand both which factors influenced the decision and why they were considered important. The fixed five-criterion schema and the JSON output format play a dual role. First, they support reproducibility and auditing: every evaluation produces a uniformly structured record in which the criterion outcomes, the assigned weights, the aggregated score, and the textual justification are stored in named fields, allowing independent reviewers to recompute aggregate scores, compare weight distributions across models or runs (as in Tables 1 and 2), and trace any narrative claim back to a specific criterion-level input. Second, they support pedagogical reuse: because the same five criteria and the same output fields recur for every message, the schema doubles as an instructional scaffold that learners and teachers can apply to new SMS examples without re-engineering the analysis, turning each output into a worked example that follows a predictable template. In this sense, the structure is not only explainable; it is also reusable for instruction and auditing. This level of explainability and structural reusability enables pedagogical recontextualization within educational settings, as it makes visible the systematic logic by which risks are evaluated. The open availability of the underlying artifacts further supports reproducibility. The curated domain allowlist, the URL shortener list, the labeled Lithuanian SMS reference corpus with its precomputed embeddings, and the search-engine-indexing script are released in the public project repository (https://github.com/LU-IZPF/smishing).

This distinction is particularly significant in educational contexts.

Developer-level interpretability supports debugging and system validation, but user-facing explainability supports learning. A technically transparent system does not automatically foster cybersecurity literacy; a system that communicates its reasoning in cognitively accessible language can actively scaffold critical evaluation skills. In this sense, the present prototype aims to transform detection from a purely protective mechanism into an instructional resource.

By incorporating explainability, our prototype aligns with calls from the XAI community to produce AI systems that support cognitive learning. Like the EXPLICATE framework, which employs LIME and SHAP to ensure explanation consistency and translate technical insights into layperson-accessible narratives (Lim et al., 2025), our system generates a structured JSON report detailing the rationale behind each risk score. This transparency fosters both trust and comprehension - two pillars emphasized in educational cybersecurity research (Melo et al., 2022).

Moreover, this approach directly addresses the findings of Timko and Rahman (2023), who reported that many anti-smishing tools either over-flag benign content or fail to inform users why a message is risky. Our explainable prototype, particularly through feature weighting and justification, functions not only as a detector but also as a tutor.

The comparative evaluation revealed substantial differences among LLMs in scoring consistency. Claude Sonnet 3.7 exhibited the highest stability (avg. CV = 0.0975, high-stability band), whereas GPT o1, despite high average scores, showed more volatility, particularly in borderline or benign messages. DeepSeek V3 provided a balanced profile, often assigning lower scores when domain indexing and allowlist checks were positive. Although the structured JSON output, comprising a narrative report, criterion weights, and an overall score, is designed to support interpretability as well as auditability and instructional reuse - the same fields that make an output legible to a learner also make it parseable for reviewers seeking to verify weight allocations or reconstruct the score - a systematic evaluation of explanation quality was beyond the scope of this study.

Importantly, the evaluation was designed to span three distinct LLM ecosystems rather than three variants from a single model family. The tested models originate from three independent providers - OpenAI (GPT o1), Anthropic (Claude Sonnet 3.7), and DeepSeek (DeepSeek V3) - each developed with its own training data, post-training procedures, and alignment strategies. Despite these differences in provenance, all models were evaluated under an identical pipeline: the same OCR-extracted inputs, the same five-criterion prompt schema (Appendix A), the same retrieval components, and the same number of repetitions per message. Holding the pipeline constant while varying the underlying LLM ecosystem is methodologically important: the stability ordering (Claude Sonnet 3.7 < DeepSeek V3 < GPT o1 in CV) and the convergent prioritization of semantic similarity as a top-ranked criterion cannot be plausibly attributed to a single provider's design choices, training corpus, or alignment strategy. This cross-ecosystem consistency strengthens the external validity of the stability and weighting observations. It provides empirical guidance for selecting language models to pedagogically recontextualize them in future educational cybersecurity systems, where reliability and reproducibility are essential. These variations reflect differing internal optimizations and prompt-following behaviors across model architectures, suggesting that model selection is crucial depending on whether the primary objective is detection accuracy or educational interpretability. Notably, semantic similarity received the highest weight in the cross-model average (23.5%) and ranked first or second for every model tested. This is significant because semantic similarity is the only criterion that evaluates message content rather than URL metadata. The remaining four criteria – search engine indexing, domain allowlist verification, URL shortener detection, and phishing database matching – all operate on URL-level signals that are unavailable when a message contains no link or when an attacker registers a previously unseen domain. Semantic similarity, by contrast, can flag a novel message if its linguistic pattern closely matches known phishing examples in the reference corpus, offering a degree of resilience against zero-day URLs that static lists cannot cover (Ghalechyan et al., 2024). The fact that all three LLMs independently assigned it the highest or near-highest weight, without explicit instruction, suggests that embedding-based reasoning is a particularly informative signal for phishing risk assessment, consistent with recent comparative work (Alhuzali et al., 2025).

Drawing on Bernstein's theory of pedagogic recontextualization (1996), the smishing risk detection model is repositioned from cybersecurity practice to pedagogic recontextualization. By integrating explainable AI components, the model's decision logic becomes visible and pedagogically accessible, enabling it to transform from a technical classification tool into an epistemic resource for developing cybersecurity literacy.

Integrating cybersecurity tools into education requires not only technical accuracy but also usability, clarity, and pedagogical recontextualization. RAG systems, when combined with a structured explanation format, fulfill these needs. Classrooms can use them as demonstrative tools to foster discussions about phishing cues and response strategies.

Additionally, research by Edwards et al. (2023) highlights a persistent gap between user knowledge and behavior, particularly in the context of SMS. Even when aware of the risks, users still click unknown links because of cognitive shortcuts and misunderstandings of cues. Our prototype aims to bridge this behavioral-attitudinal gap by promoting not only awareness but also critical analysis. It is designed to address two long-standing cybersecurity challenges simultaneously: the black-box nature of high-accuracy machine learning models and the persistent gap between behavior and attitude in user awareness. By combining retrieval-based evidence with user-facing natural-language explanations, the system transforms opaque automated judgments into transparent reasoning processes that not only detect smishing but also support critical evaluation and behavioral reflection. From the perspective of Bernstein's pedagogic recontextualization (Bernstein, 1996), the structured five-criterion output can be interpreted as a technical form that may later be transformed into pedagogic discourse. However, such transformation remains prospective and would require empirical evaluation with learners and educators. These tools also have the potential to empower teachers. As McLeod et al. (2024) show, educators struggle to access useable, trustworthy cybersecurity content. By offering transparent, explainable output, our tool can serve as both curriculum content and an instructional aid, aligning well with frameworks such as UNESCO's Digital Literacy Global Framework (Buchan et al., 2024).

From a deployment perspective, several pipeline stages (e.g., search engine indexing verification, domain allowlist lookup, phishing database check, and embedding computation) are independent. They can be executed in parallel, thereby substantially reducing end-to-end processing time. Furthermore, the rapidly evolving open-source LLM ecosystem offers increasingly viable alternatives to the commercial API-hosted models used in this study, potentially reducing costs and external dependencies while enabling local deployment.

While prior work, such as EXPLICATE (Lim et al., 2025), demonstrates the feasibility of combining high detection accuracy with explainable outputs in email-phishing contexts, the present study extends this paradigm in several important ways. First, the focus on smishing rather than email phishing introduces distinct structural and contextual constraints. SMS messages are significantly shorter, often contain minimal metadata, and frequently rely on compact linguistic manipulation or shortened URLs. Unlike email environments, SMS interfaces provide limited sender information, no visible headers, and reduced contextual cues. Consequently, detection strategies developed for email cannot be directly transferred without adaptation.

Second, the integration of an RAG pipeline tailored to SMS-specific signals, including search engine indexing checks, curated domain allowlists relevant to Lithuanian institutions, URL shortener detection,

and semantic similarity to localized smishing corpora, represents a domain-specific architectural contribution and a deliberate response to a low-resource language setting. Where large, labeled smishing corpora are unavailable, retrieving evidence at inference time and reasoning over similarity to a small, curated reference set is a more pragmatic strategy than training a supervised classifier from limited labeled data, and the same design is transferable to other low-resource languages by replacing the reference corpus and the curated lists. Rather than applying a generic XAI wrapper to a classifier, the system embeds retrieval mechanisms directly into the reasoning process, making external verification signals part of the explanation itself.

Third, and most importantly, the prototype explicitly frames explainability as an educational instrument. Whereas prior XAI phishing frameworks primarily emphasize trust calibration and technical transparency, the present work positions user-facing explainability as a foundation for digital risk literacy in mobile communication contexts. In doing so, it shifts the objective from post-hoc model justification toward cognitively accessible risk reasoning adapted to smishing's unique characteristics.

Together, these distinctions underscore that the contribution is not merely the application of XAI to phishing detection, but the development of a smishing-specific, RAG-driven, and education-oriented explainable framework.

6. Limitations

While this study presents promising results and introduces a novel framework for smishing detection with educational potential, several limitations constrain the framework's generalizability and practical applicability.

The evaluation was conducted on a curated set of 20 SMS messages evenly divided between phishing and benign examples. This size limits the ability to draw robust conclusions about real-world detection accuracy, and the results should therefore be interpreted primarily as evidence of scoring stability and consistency under this sample rather than as definitive performance benchmarks. Although this enabled controlled experimentation and comparative analysis across language models, the dataset remains modest. The 95% confidence intervals reported in Section 4.1 quantify the scoring precision achievable with 10 repetitions; wider intervals for certain benign messages (e.g., B5, B6, B8) reflect genuine scoring instability. Moreover, the linguistic diversity of the messages may not fully reflect the variability of smishing messages encountered across different languages, cultural contexts, and user demographics. Consequently, the system's performance in multilingual or non-standard linguistic settings requires further validation. Additionally, the domain allowlist contains 30 entries, the URL shortener list 657 entries, and the semantic reference corpus a limited number of labeled messages, all curated by the authors for Lithuanian-language SMS. These small, manually maintained datasets may not generalize to other languages, regions, or evolving attack patterns, and their completeness has not been independently validated.

Second, the semantic similarity threshold and dynamically assigned criterion weights were determined heuristically rather than through systematic optimization or sensitivity analysis. The similarity threshold was empirically fixed at 0.65 based on observations of the current reference corpus, and different threshold values could alter which reference messages are retrieved and subsequently influence LLM-generated risk assessments. Because the semantic similarity component was calibrated on Lithuanian SMS embeddings, its suitability for other languages, larger corpora, or evolving phishing strategies remains uncertain.

A related limitation concerns the temporal stability of the semantic reference database. The system relies on a static corpus of labeled Lithuanian SMS embeddings, whereas smishing tactics evolve rapidly and continuously introduce new linguistic patterns and social engineering themes. Without regular updates, data drift may reduce the relevance of cosine similarity comparisons and decrease sensitivity to novel attacks. Since the current update process depends on manual collection and labeling, maintaining the corpus introduces a continuing operational burden and adaptation delay.

The prototype was also not evaluated against adversarial evasion techniques, including homoglyph substitution, invisible Unicode characters, or deliberately malformed image-embedded text. Such techniques may circumvent OCR extraction or embedding-based semantic analysis and are increasingly observed in contemporary phishing campaigns. Therefore, framework's robustness under adversarial conditions remains unknown.

Although the framework was designed for both smishing detection and user education, the present study focused primarily on technical performance and interpretability. The educational impact on user comprehension, retention, and behavioral change was not empirically assessed, and the clarity and usefulness of the generated explanations were likewise not formally evaluated.

Another limitation arises from the probabilistic nature of LLM-generated explanations. While the risk score itself is derived from structured and verifiable inputs, the accompanying narrative explanation may contain inaccuracies, exaggerations, or unsupported inferences. Such hallucinated reasoning was not systematically evaluated in this study and may be particularly problematic in educational contexts if users interpret explanations as authoritative. Consequently, explanation faithfulness and factual alignment require further investigation.

The prototype's multi-stage pipeline – encompassing OCR, search engine indexing verification, embedding generation, and LLM inference – relies on multiple API calls, which introduces cumulative latency and per-token processing costs that may limit scalability in real-time deployment scenarios compared to lightweight, on-device classifiers such as the one proposed by Seo et al. (2024).

7. Conclusion

The collected data allow us to conclude that the answers to the research questions are summarized below.

RQ1: Claude Sonnet 3.7 exhibited the greatest consistency among the tested LLMs, with the lowest mean coefficient of variation (0.0975) and the highest intra-class correlation coefficient (ICC = 0.980, 95% CI [0.964, 0.991]). These results indicate notably lower variability in phishing risk scores compared to GPT o1 (ICC = 0.920) and DeepSeek V3 (ICC = 0.935), underscoring its reliability in providing stable, reproducible outcomes essential for practical applications.

RQ2: All three models consistently identified general risk trends, assigning higher scores to phishing messages and lower scores to benign ones. However, they diverged most on messages with mixed risk signals, where some criteria indicated risk while others did not. GPT o1 was the most sensitive, tending to assign elevated scores even to benign messages, while DeepSeek V3 tended to assign lower scores when domain-level checks were positive. Claude Sonnet 3.7 produced the most conservative and reproducible scoring profile across both message types.

RQ3: GPT o1 assigned the highest weight to semantic similarity (27.4%), followed by domain allowlist verification (21%). DeepSeek V3 prioritized domain allowlist verification (25%) and search engine indexing (22%), while assigning comparatively less weight to semantic similarity (20.6%). Claude Sonnet 3.7 assigned the highest weights to domain allowlist verification (23%) and semantic similarity (22.4%) as its top two criteria. All three models consistently assigned the lowest weight to URL shortener detection (~15%).

RQ4: The design attributes of the prototype possess the necessary technical attributes (transparency, consistency) to serve as a candidate for an instructional tool with the potential to support critical

thinking and provide a detailed understanding of cybersecurity threats. The fixed five-criterion structure and its machine-parseable output are not only explainable to end users; they also support reproducibility and external auditing of risk decisions and provide a stable template that can be reused across messages and teaching sessions. Reproducibility and external auditing are operationalized through the public release of the supporting artifacts (the curated domain allowlist, the URL shortener list, the labeled Lithuanian SMS reference corpus with precomputed embeddings, and the search-engine-indexing script) in the project repository (https://github.com/LU-IZPF/smishing), which lowers the barrier for independent verification, replication, and extension of the evaluation By enabling explainability in AI-driven risk assessment, the proposed tool provides a basis for pedagogical recontextualization, transforming professional cybersecurity knowledge into a form that can be meaningfully integrated into educational contexts in future applications to support the development of cybersecurity literacy.

Although the prototype is intentionally designed with interpretability features aligned with principles of cybersecurity education, the present investigation does not include an empirical assessment of pedagogical outcomes, such as learner comprehension, behavioral change, or instructional usability. Consequently, future research must empirically evaluate the educational effectiveness of the proposed system. From a hermeneutic perspective, the explainability of AI-mediated risk assessment has the potential to enable learners to engage interpretively with the system's reasoning, fostering meaning-making rather than passive acceptance of algorithmic outputs. The results instead establish a technically robust, interpretable foundation on which future empirical studies can examine user learning outcomes, classroom integration strategies, and measurable improvements in digital risk literacy.

Future research should therefore include controlled educational trials on how learners engage dialogically with AI-generated explanations to construct meaning rather than passively receive algorithmic outputs. Together, these perspectives frame explainability not only as a structural condition for the pedagogical relocation of expert knowledge but also as an epistemic space in which understanding is actively negotiated, thereby supporting the potential development of cybersecurity literacy, user-centered evaluation of explanation clarity, and longitudinal assessment of behavioral impact to determine whether the prototype substantively improves smishing detection skills and cybersecurity awareness among learners.

CRediT authorship contribution statement

Daiga Kalniņa: Writing – review & editing, Writing – original draft, Conceptualization. **Edgaras Norgaila:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Linda Daniela:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ssaho.2026.103485.

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